EECE 571Q

Lecture 14: Decoding

Tuesday 14 July 2026

Lecture Outline

- **Review of decoding objective**
- **Classical decoding algorithms**
  - Union Find
  - MWPM and the Blossom algorithm
  - Belief Propagation w/ OSD and LSD
- **Real Time Decoding**
  - Requirements for RT decoding
  - Decoder error models
  - Performance metrics
- **Demonstration**

Learning Outcomes

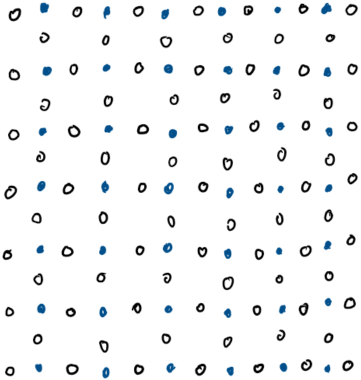
- **LO1:** Formulate quantum decoding as an optimization problem over recovery operations
- **LO2:** Explain why maximum-likelihood decoding (MLD) is generally impractical and approximate decoders are needed
- **LO3:** Apply the algorithms of Belief Propagation (BP) and its variants, Minimum Weight Perfect Matching (MWPM) and Union-Find (UF), from syndrome to recovery
- **LO4:** Identify common performance metrics in QEC decoding, comparing algorithms and adapting to face the challenge of real-time decoding

Last time

- Tanner Graphs
- Belief Propagation + Ordered Statistical Decoding
- Minimum Weight Perfect Matching

Maximum likelihood decoding

Choosing error with the highest probability of occurring:



What can we do?

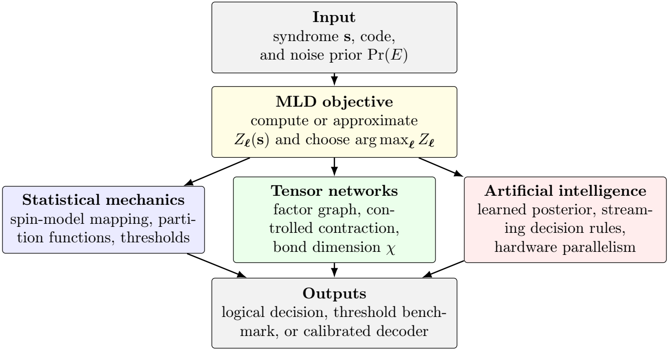
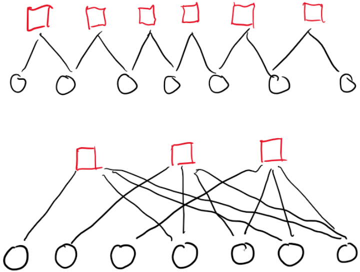


Figure: A classification of the recent approaches to quantum decoding adapted from [Cao, Yan, Du, Pan, 2026]

Tanner Graphs



Union-Find

Finds paired error syndromes on topological codes.

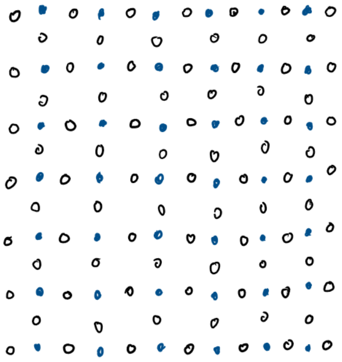
Steps:

1. Initialize clusters at syndrome defect locations
2. Grow clusters outward by $1/2$ edge at each step
3. Check parity, odd clusters have an unresolved error chain
4. Apply peeling algorithm finding a path between clusters and syndrome nodes

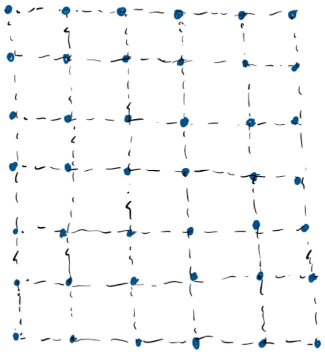
Will give a perfect matching, not necessarily minimum weight

Cluster connecting both boundaries can have a logical error

UF Example

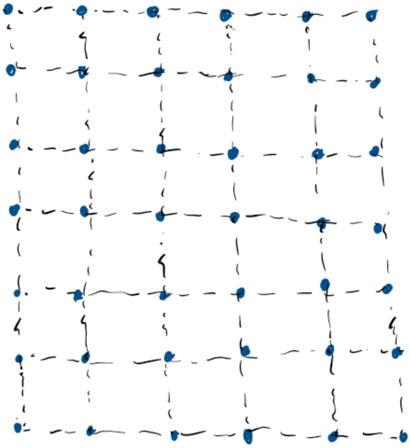


d = 7 Surface Code w/ Z stabilizers



Union Find Setup

UF Example Ctd.



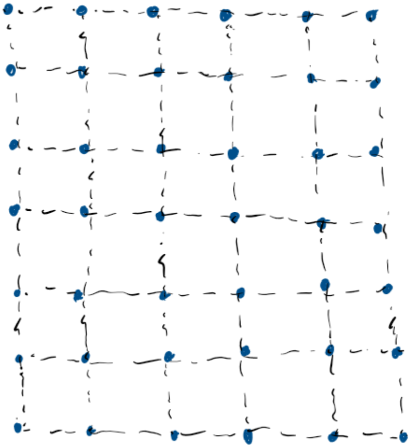
MWPM and the Blossom algorithm

Finds minimum weight paired error syndromes on topological codes.

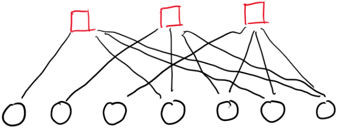
Steps:

1. Same setup as UF
2. If blossoms merge into odd cluster, begin a new blossom
3. Matches stop expanding
4. If another blossom merges with match, apply alternating tree
5. Blossoms terminate when even parity or border touched
6. Matching can only occur within the same blossom

MWPM Example



Belief Propagation (review)



BP Steps

1. Initialize data bits with p
2. Error probability on physical bits passed to check nodes
3. Beliefs updated using syndrome and sum-product rule
4. Beliefs passed back to the data bits
5. Hard decision made on updated beliefs, check if $He_{BP} = s$

OSD (review)

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix}$$

OSD Steps

1. Sort and reorder matrix by BP reliability
2. Gaussian Elimination to get pivots and free set
3. Solve for pivot bits by assigning hard decisions to free bits
4. Enumerate and score bit flip combinations in free set
5. Output lowest cost candidate error pattern

BP and Localized Statistics Decoding

OSD's major bottleneck is the global matrix solve
We need a way of limiting this overhead by parallelizing this step

Idea - Only solve local parity check matrices using OSD, treat as clusters
Then we can parallelize each cluster

LSD: Local Matrix Solves

Global decoding problem: $He = s$

LSD separates this into local clusters which can be solved in parallel

$$H_{C_i} e_{C_i} = s_{C_i}$$

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} e_3 \\ e_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

LSD: On-the-Fly Elimination

1. Initial C_1

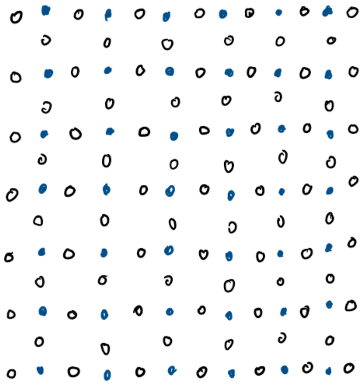
$$\left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{array}\right]$$

2. Grow C_1

$$\left[\begin{array}{ccc|c} 1 & 0 & \color{red}{0} & 1 \\ 0 & 1 & \color{red}{0} & 0 \\ 1 & 1 & \color{red}{1} & 0 \end{array}\right]$$

$$e_1 = 1, \qquad e_2 = 0, \qquad e_3 = 1$$

Localized Statistics Decoding



Interlude

Questions?

All references will be included in lecture notes, as well as some additional readings on some modern approaches to these algorithms such as fusion and sparse blossoms for MWPM

Requirements for real time decoding

If $r_{gen}/r_{proc} = f > 1$, a small initial backlog in processing syndrome data will lead to an exponential slow down during the computation. [Terhal, 2015]

Detector Error Models

Circuit / Hardware

$\downarrow$

Detector Error Model (DEM)

$\downarrow$

Decoder

- Bridge circuit level noise and decoder operation
- Probabilities associated with combinations of phsyical faults
- Encodes all this information and passes to decoder

Performance Metrics

Metrics used to evaluate QEC Decoders in practice:

- Error threshold t
- Logical Error rate p_l
- Decoding Time (per cycle) T_d

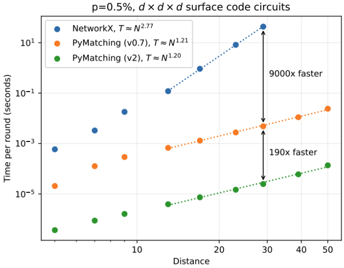


Figure: A plot of decoding time per cycle against distance for pymatching [Sparse Blossom: correcting a million errors per core second with minimum-weight matching, Higgott, Gidney, 2025]

Demo 1

Comparison of various decoders on different codes

Showcasing performance and capabilities of different decoders in different situations